Cafe Sales Performance and Customer Behaviour Analysis

Sector Name: Food & Beverage / Retail Analytics.

Team ID : Group-16

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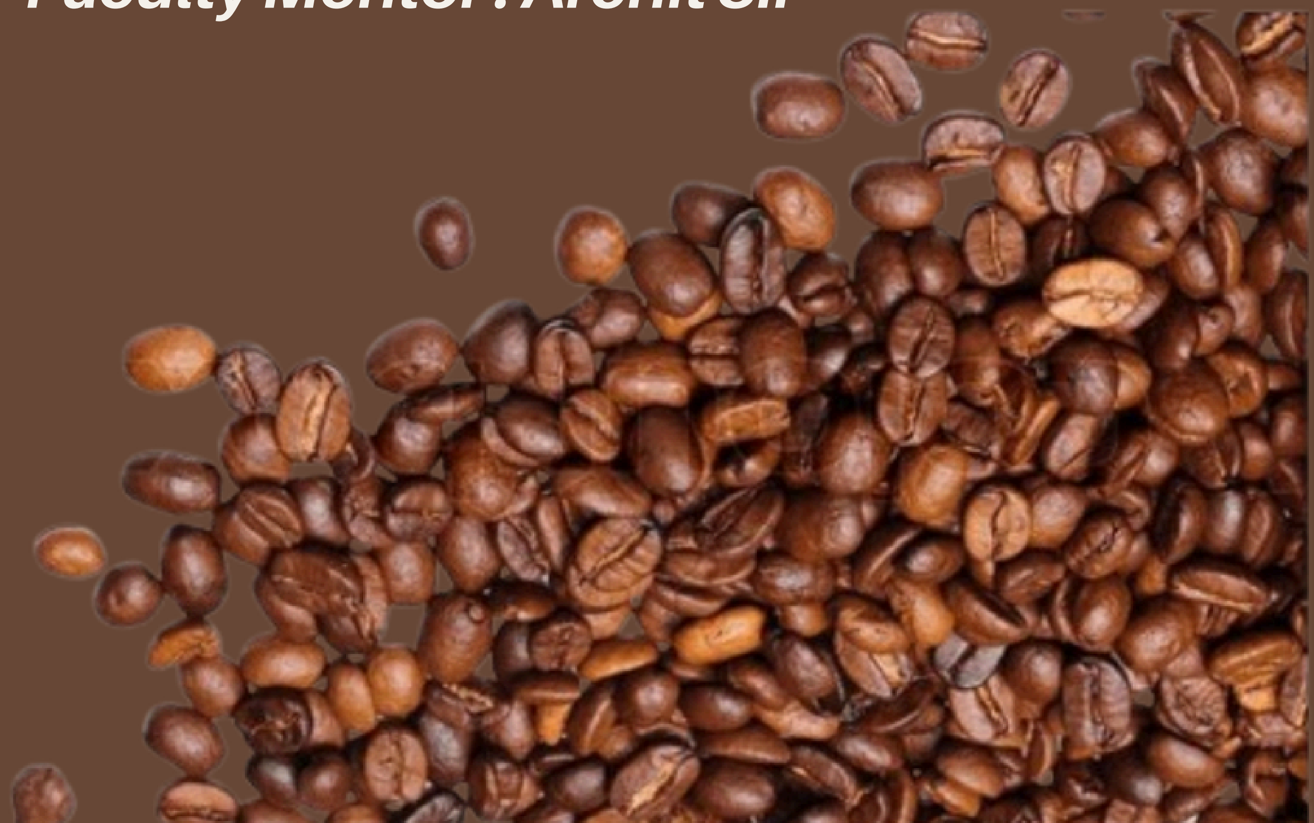


Context and Problem Statement

In a cafe, managing fresh food and drinks is critical because items like sandwiches and smoothies can spoil. Decisions are based on which days of the week or months are the busiest

How can we manage our food stock better when the sales records have errors and missing information?

HOW WE WILL SOLVE THE PROBLEM?

DATA INTEGRITY

To rectify calculation errors and reconcile missing transaction details.

STOCK OPTIMIZATION

To identify high-volume sales dates to ensure optimal food and beverage stock levels.

STRATEGIC PLANNING

To provide a reliable baseline for menu pricing and payment processing decisions.

Data Engineering

SOURCE

Name
Cafe_Sales.csv;
Rows/Cols
7774 transactions;
Time Period
Full Year 2023.

CLEANING

MISSING VALUES
Rows with missing/"ERROR" in critical fields removed. Calculation integrity ensured.
STANDARDIZATION
Ambiguous/empty Payment Method/Location grouped into "Unknown."
DATA CONVERSION
Quantity, Price, Total Spent to numeric; Transaction Date to Datetime.
Mathematical Validation
"Total Spent" recalculated (Quantity × Price) to eliminate entry errors.
FINAL VERDICT
Comprehensive check confirms 7,774 transactions meet type/completeness standards for analysis.

DICTIONARY

- Transaction ID
- Item
- Quantity
- Price Per Unit
- Total Spent
- Payment Method
- Location
- Transaction Date

KPI & Metrics Framework

Total Revenue	Total Quantity	Total Transactions
$69459	23505	$7773

Linking Metrics to Business Model:

- **Solving the Revenue** Leakage: By measuring True Gross Revenue, we address the problem of underreporting caused by the "ERROR" logs in the raw data.
- **Waste Reduction**: Volume by Item directly links to our problem statement of managing perishable stock.
- **Resource Allocation**: By measuring Channel Contribution, the decision maker can decide whether to spend more on "Takeaway" packaging or "In-store" seating and staffing.

Key Insights (EDA)

Strategic Observations

- Coffee remains the most frequent purchase, indicating high customer retention and daily habits.
- Salad contributes 21.4% of total revenue, indicating it is more expensive than rest of the items.
- Low-cost items like Cookies are consistently paired with larger orders.
- October and June are the highest performing periods.
- Smoothie demand increased by 70%, whereas Sandwich demand decreased by 20% between January and June.
- Sales are nearly perfectly balanced between In-store and Takeaway, requiring a Hybrid service focus.
- Digital Wallet and Credit Card payments are now equal to Cash, suggesting a shift toward a cashless model.

Advanced Analysis

Seasonal Variance Analysis:

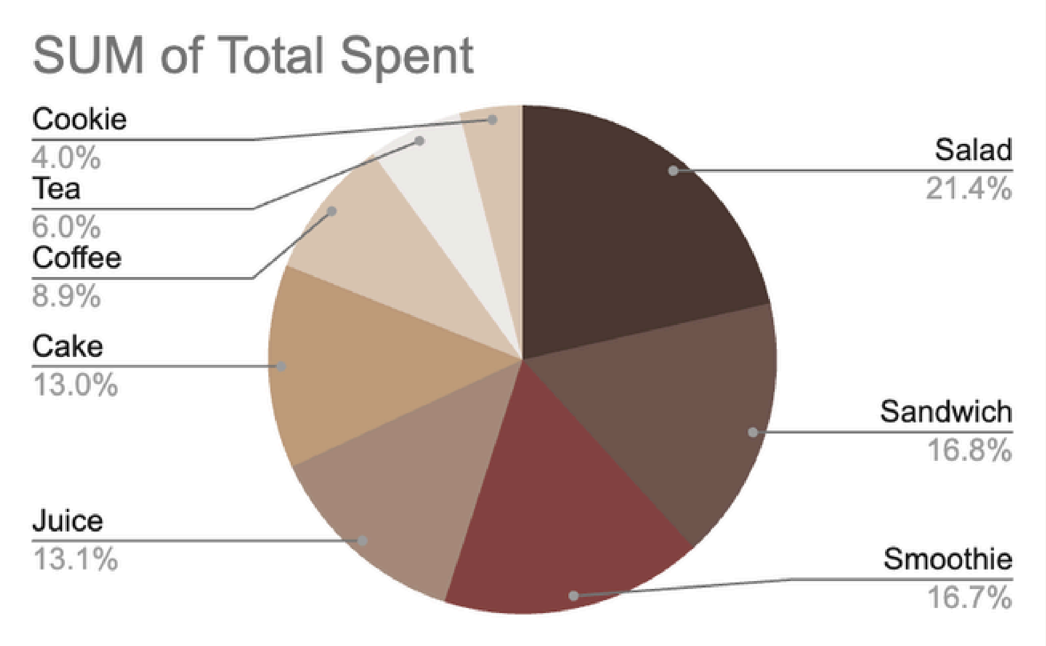
- A comparative study between January and June reveals a 70.4% surge in Smoothie demand, indicating high sensitivity to summer temperature peaks.
- Conversely, "Heavy" items like Sandwiches and Cakes saw a 15-20% decline during the same period, suggesting a seasonal shift in consumer appetite.

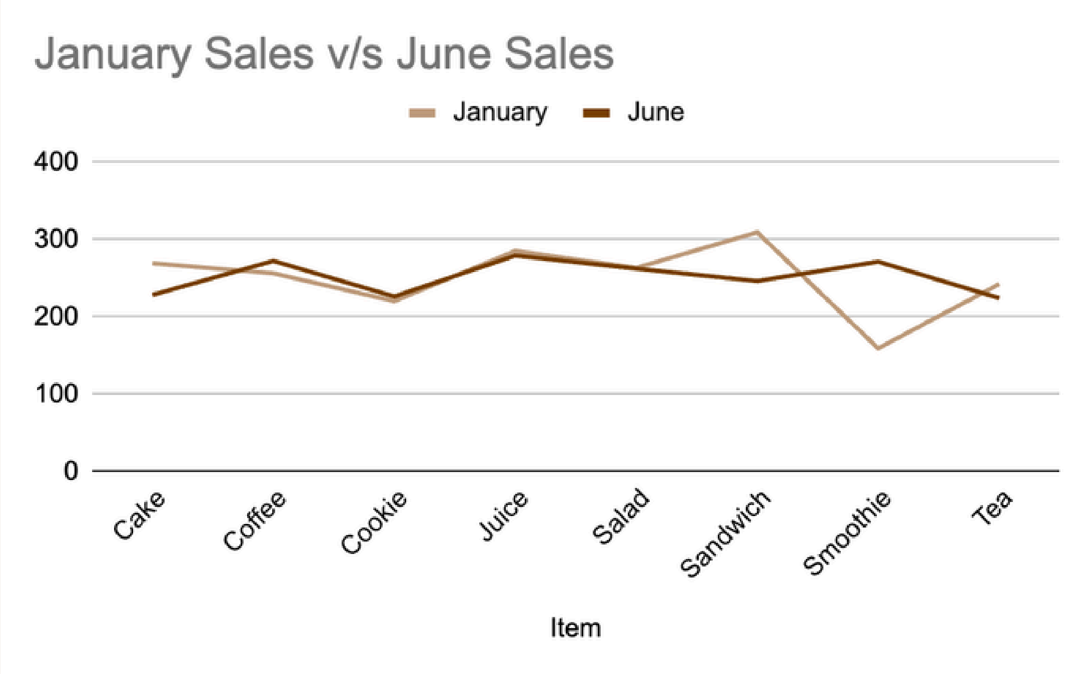
Location-Menu Affinity:

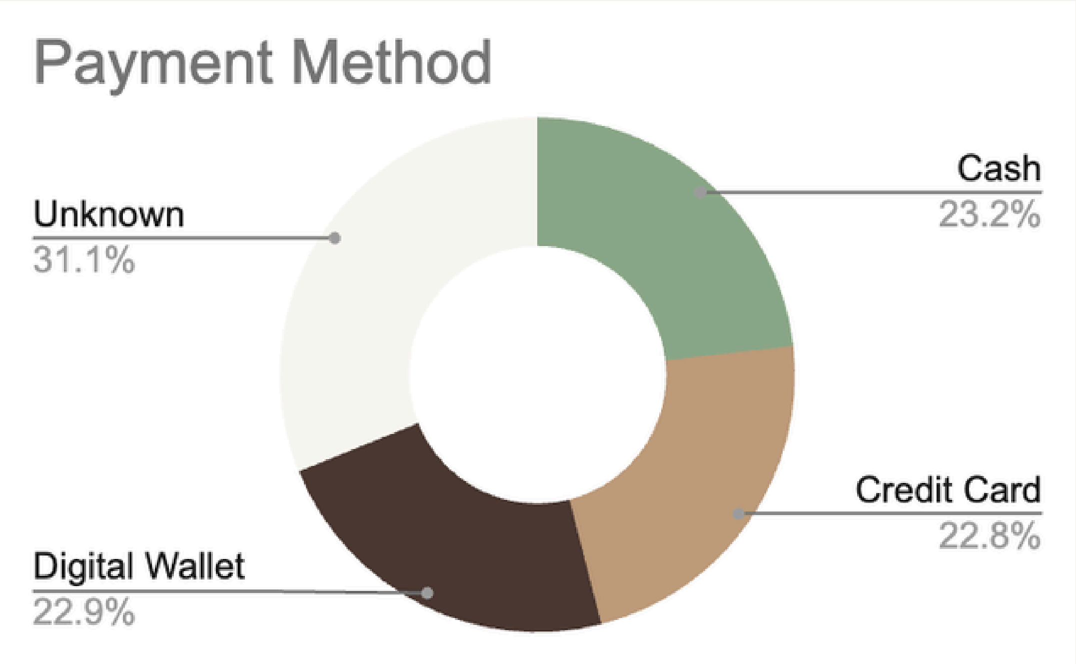
- Takeaway orders are dominated by Quick Consumables like Coffee and Cookies (900+ units each)
- In-store dining shows a higher preference for "Meal-based" items like Salads and Juices.

Systemic Data Gaps:

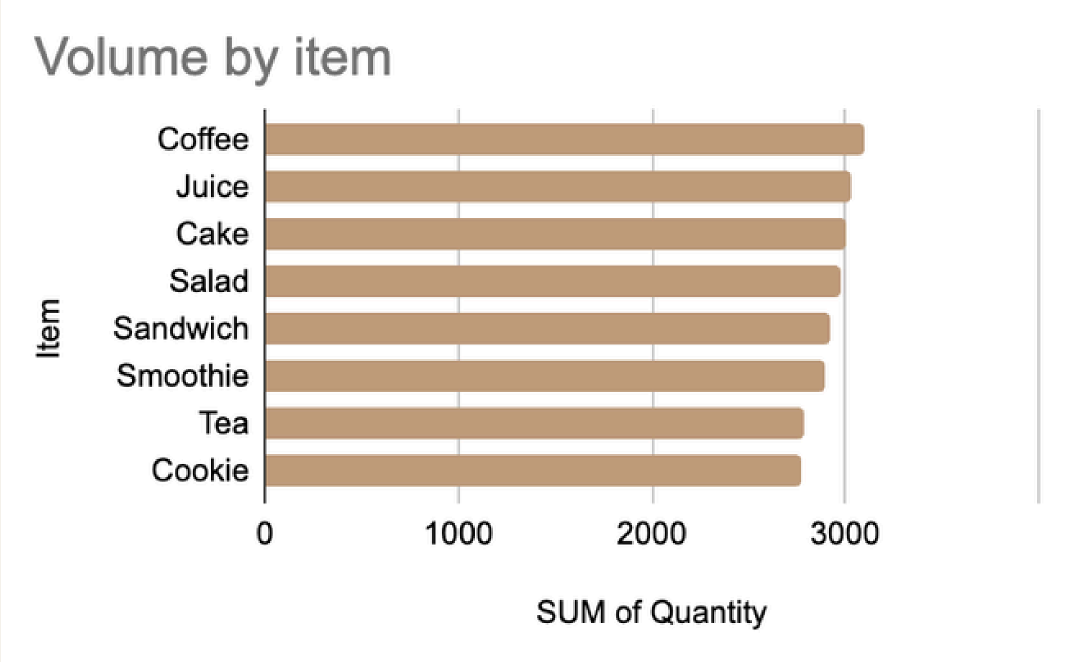
- The 31.6% "Unknown" payment rate shows no seasonal pattern, confirming it is a year-round technical glitch rather than a human error.



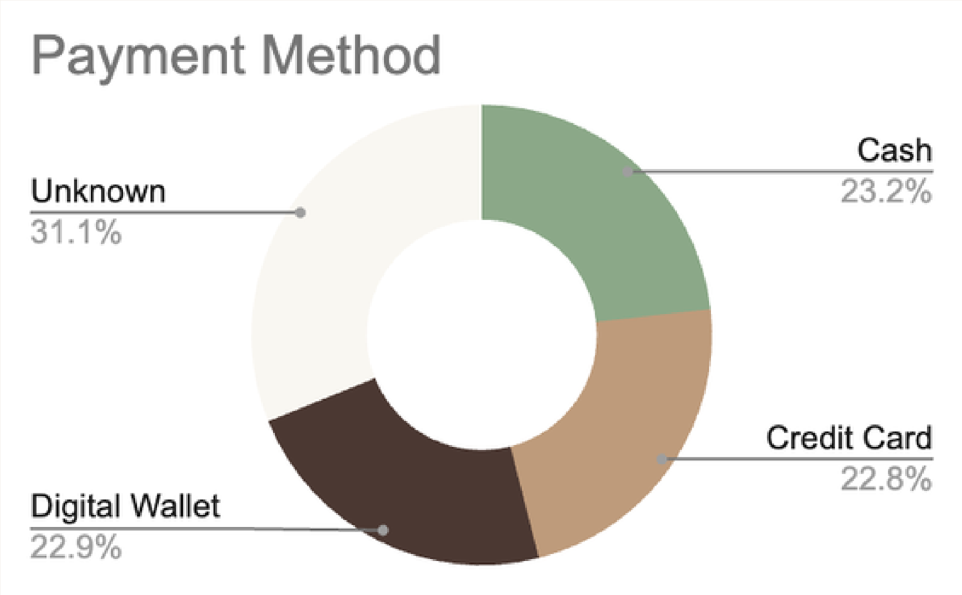




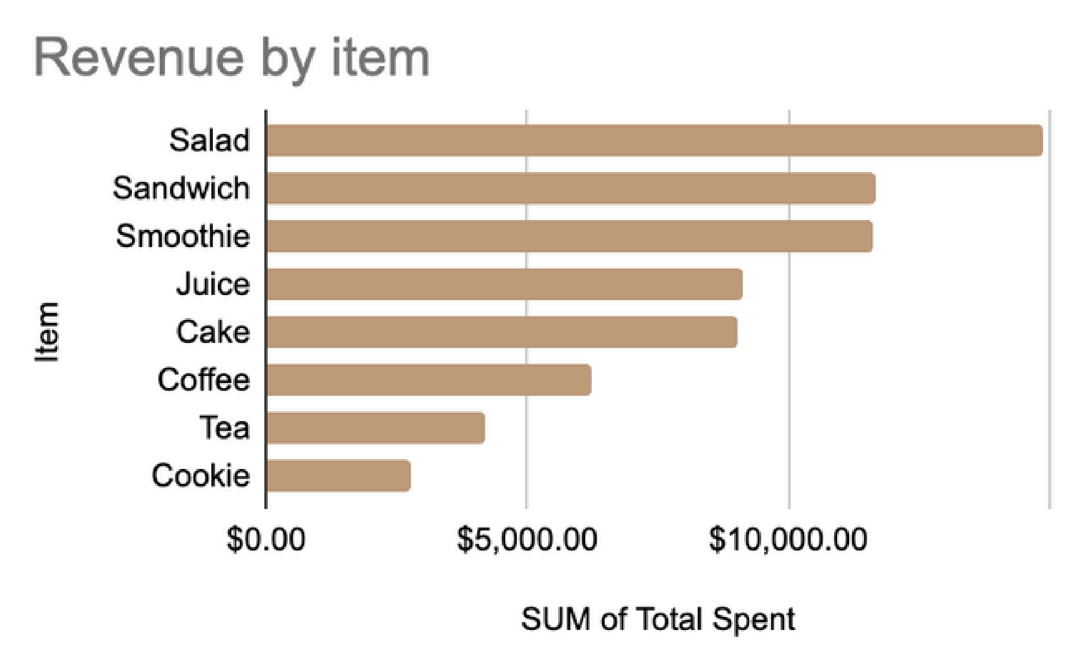
Dashboard Walkthrough



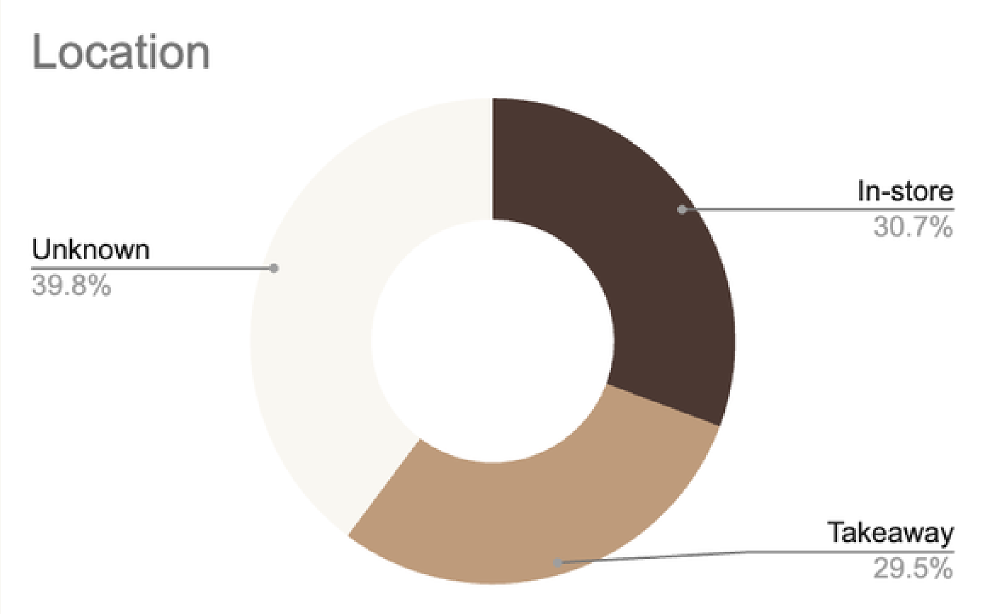
- Coffee, Juice, and Cake are the most frequently ordered items.
- Sandwiches and Smoothies also show strong, consistent demand across all months.



- Cash, Credit Card, and Digital Wallets are used almost equally (~23% each).
- 31.1% of transactions are marked "Unknown," highlighting a critical need for system reconciliation.



- Salads and Sandwiches generate the highest total revenue.
- Monthly revenue remains stable at approximately $6,000, totaling $69,459 annually.



- Orders are evenly split between In-store (30.7%) and Takeaway (29.5%).
- A significant 39.8% of orders have an "Unknown" location, hindering logistics planning.

Recommendations

Fix Technical Issues (Payments & Tracking)

Fix the system to tag every transaction with a User ID so we don't have UNKNOWN payments anymore.

Improve the order delivery status so that we know whether it's a Takeaway or Dine-In.

Season Matching Management

Keep supplies ready during trendy months to meet high demand.

Better Seating & In-Store Experience

Optimize Layout: Rearrange the tables to create more space for "in-place" diners.

Dedicated Waiting Zone: Set up a small area for "to-go" orders so people waiting don't block the people trying to eat.

Loyalty & Feedback

Instant Feedback: Put a QR code on tables so customers can tell us immediately if the seating or quality isn't up to mark.

Impact & Value

Revenue & Growth

By maintaining season specialist stock, we can prevent stock-outs and capture a 20-30% increase in seasonal revenue.

The loyalty program and QR feedback will help convert one-time visitors into regulars, increasing long-term sales.

Operational Efficiency

Improving the seating layout and adding a waiting zone will speed up the flow of both Dine-In and Takeaway orders.

Proper season management ensures staff spend less time dealing with shortages and more time serving customers.

Accuracy & Trust

Fixing the UNKNOWN payments issue ensures every transaction is tracked, eliminating revenue leakage and accounting errors.

Clearer order status reduces confusion for the kitchen staff and ensures customers get their orders exactly how they want them.

Limitations & Next Steps

Limitations

- Current data doesn't show exactly why payments are failing, only that they are "UNKNOWN.”
- The physical size of the shop might limit how many new seats we can actually add.
- Summer demand can fluctuate based on heatwaves or rain, making stock management tricky.

Next Steps

- Incorporate cost and margin data to enable profitability and contribution margin analysis.
- Improve POS data validation to eliminate missing location and payment entries.
- Apply forecasting techniques to predict future sales and support inventory and staffing planning.
- Extend the analysis using advanced visualization or BI tools for enhanced interactivity and scalability.